

COMPANION FOR CHAPTER 5

OF

**FUNDAMENTALS
OF PROBABILITY**
WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 5

Special Discrete Distributions

5A ADDITIONAL EXAMPLES

Example 5a A realtor claims that only 30% of the houses in a certain neighborhood are appraised at less than \$200,000. A random sample of 20 houses from that neighborhood is selected and appraised. The results in (thousands of dollars) are as follows:

285	156	202	306	276	562	415
245	185	143	186	377	225	192
510	222	264	198	168	363	

Based on these data, is the realtor's claim acceptable?

Solution: Suppose that the realtor's claim is acceptable, and $p = 0.30$ is the probability that a randomly selected house is appraised at less than \$200,000. In a randomly selected sample of 20 houses, let X be the number that are less than \$200,000. Since in the data given there are seven such houses, we will calculate $P(X \geq 7)$. If this probability is too small, we will reject the claim of the realtor; otherwise, we will accept it. To find $P(X \geq 7)$, note that X is binomial with parameters 20 and 0.30. Therefore,

$$P(X \geq 7) = 1 - P(X \leq 6) = 1 - \sum_{i=0}^6 \binom{20}{i} (0.30)^i (0.70)^{20-i} \approx 0.392,$$

which shows that the event that seven or more houses are appraised at less than \$200,000 is highly probable. So the claim of the realtor should be accepted. ♦

Example 5b For a manufacturer of Go-Into-Location-in-Space (GOLIS) missiles, there are two companies that build missile trackers for guided missiles that are designed for moving targets. The manufacturers purchase 40% of their missile trackers from company I and the remaining 60% from company II. Suppose that 8.7% of the missile trackers designed by company I and 7.3% of the missile trackers designed by company II are defective. In a GOLIS missile test, a batch of 100 such missiles, all equipped with missile trackers designed

by the same company, were launched attacking moving targets. If 9 of the missiles failed to reach their intended targets, all because of the defects in their missile trackers, what is the probability that the batch of 100 missiles were all manufactured by company I?

Hint: Use Bayes' formula.

Solution: Let X be the number of missiles that failed to reach their intended targets. Let A be the event that the batch of 100 missiles all had missile trackers designed by company I. We are interested in $P(A \mid X = 9)$. By Bayes' formula,

$$\begin{aligned} P(A \mid X = 9) &= \frac{P(X = 9 \mid A)P(A)}{P(X = 9 \mid A)P(A) + P(X = 9 \mid A^c)P(A^c)} \\ &= \frac{\binom{100}{9}(0.087)^9(1 - 0.087)^{91} \cdot 0.4}{\binom{100}{9}(0.087)^9(1 - 0.087)^{91} \cdot 0.4 + \binom{100}{9}(0.073)^9(1 - 0.073)^{91} \cdot 0.6} \approx 0.81. \end{aligned}$$



Example 5c In a certain population, one percent of the people are colorblind. In order to calculate the probability of exactly i colorblind people among a random sample of n , we use the binomial probability function with parameters $(n, 0.01)$ and let

$$b_i = \binom{n}{i}(0.01)^i(0.99)^{n-i}.$$

Then we use Poisson probability function with parameter $\lambda = n(1/100) = n/100$ and let

$$p_i = \frac{e^{-(n/100)}(n/100)^i}{i!}.$$

Both of these can be used only as approximations. They are not exact, since in binomial (and hence in Poisson approximations) the “trials” are independent. Here the “trials” are dependent, because the number of colorblind people in the population is fixed. Therefore, for, say, $0 < i, j < n$, the probability that the j th person in the sample is colorblind depends on whether the i th person is colorblind or is not. The probability that the j th person is colorblind is less if the i th person is colorblind than if he or she is not colorblind. However; if n is not too small, then this dependence is very weak, and both binomial and Poisson are good approximations.

For different values of n and i , the following tables show how b_i 's and p_i 's are in relative agreement.

(i) $n = 20$

i	b_i	p_i
0	.8180	.8187
1	.1652	.1638
2	.0159	.0164
3	.0010	.0011
4	.00004	.00005

(ii) $n = 30$

i	b_i	p_i
0	.7397	.7408
1	.2242	.2222
2	.0328	.0333
3	.0031	.0033
4	.0002	.0003
5	.00001	.00002

(iii) $n = 50$

i	b_i	p_i
0	.6050	.6065
1	.3056	.3033
2	.0757	.0759
3	.0122	.0126
4	.0015	.0016
5	.00013	.00016
6	.00001	.00001



Example 5d The probability of winning a certain game is p and the probability of losing it is $1 - p$. Players A and B keep playing this game until one of them wins n games. The games are independent and the person who wins n games first is the winner. Suppose that after $k + r$ ($k < n$, $r < n$) games A has won k games and B has won r games. What is the probability that A wins?

Solution: In order that A wins, he must win $n - k$ games before losing $n - r$. This happens if and only if the $(n - k)$ th win of A occurs before or at the $(n - r) + (n - k) - 1 = 2n - (r + k + 1)$ th game. This follows since if the $(n - k)$ th win for A occurs after the $2n - (r + k + 1)$ th game, then B wins more than $n - r$ games that he needs. Let X be the number of games required that A wins $(n - k)$ games. Then X is a negative binomial with parameter $(n - k, p)$. The desired probability is

$$\sum_{i=n-k}^{2n-(r+k+1)} P(X = i) = \sum_{i=n-k}^{2n-(r+k+1)} \binom{i-1}{n-k-1} p^{n-k} (1-p)^{i-n+k}. \quad \blacklozenge$$

Example 5e Peggy and Norma are two basketball players. The probability that Peggy makes a foul shot, independently of her other shots, is p . The probability that Norma makes a foul shot is also p , independently of her other shots. Peggy shoots free throws until a successful one. Then she gives the ball to Norma, who shoots free throws until she makes a foul shot too. After that they both quit shooting basketball shots. If Peggy and Norma together attempted 8 shots before quitting, for $1 \leq j \leq 7$, what is the probability that Peggy made her foul shot on her j th attempt, $j = 1, 2, \dots, 7$?

Solution: Let X be the number of times Peggy attempts until she makes a foul shot. Let Y be the number of times Norma attempts until she makes a foul shot. Since X and Y are independent and identically distributed, by symmetry, we must have

$$P(X = j \mid X + Y = 8) = \frac{1}{7}, \quad j = 1, 2, \dots, 7.$$

Here is a formal proof:

$$\begin{aligned} P(X = j \mid X + Y = 8) &= \frac{P(X = j, X + Y = 8)}{P(X + Y = 8)} = \frac{P(X = j, Y = 8 - j)}{\sum_{i=1}^7 P(X = i, Y = 8 - i)} \\ &= \frac{P(X = j)P(Y = 8 - j)}{\sum_{i=1}^7 P(X = i)P(Y = 8 - i)} \\ &= \frac{(1-p)^{j-1}p \cdot (1-p)^{8-j-1}p}{\sum_{i=1}^7 (1-p)^{i-1}p \cdot (1-p)^{8-i-1}p} \\ &= \frac{(1-p)^6 p^2}{\sum_{i=1}^7 (1-p)^6 p^2} = \frac{(1-p)^6 p^2}{(1-p)^6 p^2 \sum_{i=1}^7 1} = \frac{1}{7}. \quad \blacklozenge \end{aligned}$$

Example 5f Let X be a Poisson random variable with parameter $\lambda > 0$. Show that

(a) for $n > \lambda - 1$,

$$P(X \geq n) < \frac{n+1}{n+1-\lambda} P(X = n);$$

(b) $n > 2\lambda - 1$,

$$P(X > n) < P(X = n).$$

Solution: (a) We have that

$$\begin{aligned} P(X \geq n) &= \sum_{i=n}^{\infty} \frac{e^{-\lambda} \cdot \lambda^i}{i!} \\ &= \frac{e^{-\lambda} \cdot \lambda^n}{n!} \left[1 + \frac{\lambda}{n+1} + \frac{\lambda^2}{(n+1)(n+2)} + \frac{\lambda^3}{(n+1)(n+2)(n+3)} + \dots \right]. \end{aligned}$$

Now

$$\begin{aligned} \frac{\lambda^2}{(n+1)(n+2)} &< \frac{\lambda^2}{(n+1)^2} \\ \frac{\lambda^3}{(n+1)(n+2)(n+3)} &< \frac{\lambda^3}{(n+1)^3} \\ &\vdots \end{aligned}$$

Thus

$$\begin{aligned} P(X \geq n) &< \frac{e^{-\lambda} \cdot \lambda^n}{n!} \sum_{i=0}^{\infty} \frac{\lambda^i}{(n+1)^i} \\ &= P(X = n) \sum_{i=0}^{\infty} \left(\frac{\lambda}{n+1} \right)^i \end{aligned}$$

Now, for $n > \lambda - 1$, we have $\frac{\lambda}{n+1} < 1$, and $\sum_{i=0}^{\infty} \left(\frac{\lambda}{n+1} \right)^i$ is a geometric series that converges to $\frac{1}{1 - \frac{\lambda}{n+1}} = \frac{n+1}{n+1-\lambda}$. Therefore, for $n > \lambda - 1$,

$$P(X \geq n) < \frac{n+1}{n+1-\lambda} P(X = n).$$

(b) Note that, by the result of part (a),

$$\begin{aligned} P(X > n) &= P(X \geq n) - P(X = n) < \frac{n+1}{n+1-\lambda} P(X = n) - P(X = n) \\ &= \left(\frac{n+1}{n+1-\lambda} - 1 \right) P(X = n) = \frac{\lambda}{n+1-\lambda} P(X = n). \end{aligned}$$

This shows that $P(X > n) < P(X = n)$ if $\frac{\lambda}{n+1-\lambda} < 1$ or if $n > 2\lambda - 1$. ♦

Example 5g Suppose that, from a box containing D defective and $N - D$ non-defective items, n ($n \leq \min(D, N - D)$) are drawn at random and without replacement. Let X be the number of defective items drawn. Then X is a hypergeometric random variable.

(a) Prove the following recurrence relation for X .

$$P(X = x + 1) = \frac{(n - x)(D - x)}{(x + 1)(N - D - n + x + 1)} P(X = x),$$

where $\max(0, n - (N - D)) \leq x \leq \min(n, D)$.

(b) Using part (a), calculate the probability mass function of the number of aces in a bridge hand. A bridge hand consists of 13 randomly selected cards from an ordinary deck of 52 cards.

Solution: **(a)** Note that

$$\frac{P(X = x + 1)}{P(X = x)} = \frac{\binom{D}{x+1} \binom{N-D}{n-x-1} / \binom{N}{n}}{\binom{D}{x} \binom{N-D}{n-x} / \binom{N}{n}}$$

$$\begin{aligned}
&= \frac{\frac{D!}{(x+1)!(D-x-1)!} \cdot \frac{(N-D)!}{(n-x-1)!(N-D-n+x+1)!}}{\frac{D!}{x!(D-x)!} \cdot \frac{(N-D)!}{(n-x)!(N-D-n+x)!}} \\
&= \frac{x!(D-x)!}{(x+1)!(D-x-1)!} \cdot \frac{(n-x)!(N-D-n+x)!}{(n-x-1)!(N-D-n+x+1)!} \\
&= \frac{D-x}{x+1} \cdot \frac{n-x}{N-D-n+x+1}.
\end{aligned}$$

This establishes the relation.

(b) Let X be the number of aces in a bridge hand. Clearly, X is a hypergeometric random variable with parameters $N = 52$, $D = 4$, and $n = 13$. Let p be the probability mass function of X . Clearly,

$$p(0) = P(X = 0) = \frac{\binom{4}{0} \binom{48}{13}}{\binom{52}{13}} \approx 0.3038.$$

Now by part (a),

$$\begin{aligned}
p(1) &= \frac{13 \times 4}{1 \times 36} p(0) \approx \frac{52}{36} \times 0.3038 \approx 0.4388, \\
p(2) &= \frac{12 \times 3}{2 \times 37} p(1) \approx \frac{36}{74} \times 0.4388 \approx 0.2135, \\
p(3) &= \frac{11 \times 2}{3 \times 38} p(2) \approx \frac{22}{114} \times 0.2135 \approx 0.0412, \\
p(4) &= \frac{10 \times 1}{4 \times 39} p(3) \approx \frac{10}{156} \times 0.0412 \approx 0.0026. \quad \blacklozenge
\end{aligned}$$

Example 5h Suppose that independent Bernoulli trials with parameter p , $0 < p < 1$, are performed and X denotes the number of trials until the r th success. In this example, we explain the reason why X is called negative binomial. The advantage of this explanation is that it also enables us to calculate the expectation of X and prove analytically that, for $0 < p < 1$,

$$p(x) = P(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots,$$

is a probability mass function. In Chapter 2 of the book, the notion of $\binom{n}{r}$ is defined by $n!/[r!(n-r)!]$ for positive integers n and non-negative integers r . The same notion, $\binom{n}{r}$,

can also be defined for *real numbers* n and non-negative integers r by

$$\binom{n}{r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!},$$

which obviously agrees with $n!/[r!(n-r)!]$ in the case of positive integer n . Now let Y be the number of failures before the r th success. Then $X = Y + r$, and, for $y = 0, 1, 2, \dots$,

$$P(Y = y) = P(X = y + r) = \binom{y+r-1}{r-1} p^r (1-p)^y. \quad (1)$$

But

$$\begin{aligned} \binom{y+r-1}{r-1} &= \frac{(y+r-1)!}{(r-1)! y!} = \frac{r(r+1)(r+2)\cdots(y+r-1)}{y!} \\ &= \frac{(-1)^y [-r(-r-1)(-r-2)\cdots(-r-y+1)]}{y!} \\ &= (-1)^y \binom{-r}{y}. \end{aligned}$$

Therefore,

$$P(Y = y) = (-1)^y \binom{-r}{y} p^r (1-p)^y = \binom{-r}{y} p^r (p-1)^y, \quad y = 0, 1, 2, \dots$$

It is because of this relation that X is called “negative binomial”. Using this relation, we will now show that $p(x)$ is a probability function:

$$\begin{aligned} \sum_{x=r}^{\infty} p(x) &= \sum_{x=r}^{\infty} P(X = x) = \sum_{x=r}^{\infty} P(Y = x - r) = \sum_{y=0}^{\infty} P(Y = y) \\ &= \sum_{y=0}^{\infty} \binom{-r}{y} p^r (p-1)^y = p^r \sum_{y=0}^{\infty} \binom{-r}{y} (p-1)^y \\ &= p^r (1 + p - 1)^{-r} = 1, \end{aligned}$$

where the next to the last equality comes from calculus: If r is any real number, then for

$$|t| < 1, \quad (1+t)^r = \sum_{y=0}^{\infty} \binom{r}{y} t^y.$$

Another application of relation (1) is that it enables us to find $E(X)$ easily. Note that

$$\begin{aligned} E(X) &= \sum_{x=r}^{\infty} x p(x) = \sum_{x=r}^{\infty} x P(Y = x - r) \\ &= \sum_{y=0}^{\infty} (y+r) P(Y = y) = \sum_{y=0}^{\infty} y P(Y = y) + r \sum_{y=0}^{\infty} P(Y = y) \\ &= E(Y) + r = \frac{r(1-p)}{p} + r = \frac{r}{p}, \end{aligned}$$

where $E(Y) = \sum_{y=0}^{\infty} y \binom{-r}{y} p^r (p-1)^y = [r(1-p)]/p$ follows from a straightforward procedure similar to the one we used to calculate the expected value of a binomial random variable. Note that X is the number of trials until the r th success. Since the average number of trials until the first success is $1/p$ (the expected value of a geometric random variable with parameter p is $1/p$), we should expect that the average number of trials until the r th success be r/p . This is the intuition behind the relation $E(X) = r/p$ obtained above. In Chapter 10 of the book, we have calculated $E(X)$ using a simpler method. ♦

Example 5i (Genetics) In a population of n diploid organisms with alternate dominant allele A and recessive allele a , those inheriting aa will not survive. Suppose that, in the population, the number of AA individuals is α , and the number of Aa individuals is $n - \alpha$. Suppose that, as a result of random mating, m ($m > 2$) offspring are produced.

- (a) What is the probability that at most two of the offspring are aa ?
- (b) For $0 \leq i \leq m$, what is the probability that exactly i of the offspring are AA and the remaining are all Aa ?

Solution: Let p , q , and r be the probabilities that a randomly selected offspring is AA , Aa , and aa , respectively. Note that both parents of the offspring are AA with probability $(\alpha/n)^2$, they are both Aa with probability $[1 - (\alpha/n)]^2$, and the probability is $2(\alpha/n)[1 - (\alpha/n)]$ that one parent is AA and the other is Aa . Therefore, by the law of total probability,

$$\begin{aligned} p &= 1 \cdot \left(\frac{\alpha}{n}\right)^2 + \frac{1}{4} \cdot \left(1 - \frac{\alpha}{n}\right)^2 + \frac{1}{2} \cdot 2\left(\frac{\alpha}{n}\right)\left(1 - \frac{\alpha}{n}\right) = \frac{1}{4}\left(\frac{\alpha}{n}\right)^2 + \frac{1}{2}\left(\frac{\alpha}{n}\right) + \frac{1}{4}, \\ q &= 0 \cdot \left(\frac{\alpha}{n}\right)^2 + \frac{1}{2}\left(1 - \frac{\alpha}{n}\right)^2 + \frac{1}{2} \cdot 2\left(\frac{\alpha}{n}\right)\left(1 - \frac{\alpha}{n}\right) = \frac{1}{2} - \frac{1}{2}\left(\frac{\alpha}{n}\right)^2, \\ r &= 0 \cdot \left(\frac{\alpha}{n}\right)^2 + \frac{1}{4}\left(1 - \frac{\alpha}{n}\right)^2 + 0 \cdot 2\left(\frac{\alpha}{n}\right)\left(1 - \frac{\alpha}{n}\right) = \frac{1}{4}\left(1 - \frac{\alpha}{n}\right)^2. \end{aligned}$$

The probability that at most two of the offspring are aa is

$$\sum_{i=0}^2 \binom{m}{i} r^i (1-r)^{m-i}.$$

The probability that exactly i of the offspring are AA and the remaining are all Aa is

$$\binom{m}{i} p^i q^{m-i}. \quad \blacklozenge$$